

RESEARCH NOTE

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Leveraging nationwide epidemiological big data for diabetes modeling and disease surveillance: an innovative approach for advancing public health policy in developing countries

Hadi Hayati^{1,2*} and Razieh Askari Zahabi³

Abstract

Objective Diabetes is escalating into a global crisis, with low- and middle-income countries bearing the heaviest burden. Yet few nations openly harness nationwide health data to guide precise, evidence-based interventions. This study pioneers the large-scale integration of Iran's decade-long electronic prescription records with advanced modelling tools to illuminate national and regional dynamics in antidiabetic drug use. By fusing big data analytics with policy-driven insight, our aim was not only to expose inequities and inefficiencies but to create a transferable blueprint for equitable, guideline-aligned diabetes care worldwide.

Results Across 84 million prescriptions (2013–2023), metformin dominated treatment, but novel agents such as DPP-4 and SGLT2 inhibitors surged after 2017, reshaping the therapeutic landscape. Stark contrasts emerged: first-line metformin initiation ranged from 58.3% in underserved Sistan & Baluchestan to 91.7% in resource-advantaged Tehran; inappropriate early adoption of new agents and premature insulinisation persisted in several regions. These patterns mirror global fault lines—where innovation coexists with inequity—and underscore the urgency for adaptive, data-led policy. Our proposed multi-tier model—combining real-time digital surveillance, prescriber upskilling, and patient empowerment—offers a novel, globally relevant pathway for transforming big data into equitable, future-ready diabetes strategies in resource-limited health systems.

Keywords Big data, Epidemiology, Antidiabetic agents, Open dataset, Disease modeling, Surveillance, Non-communicable diseases, Sustainable development goals, Public health policy, Iran

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Introduction

Diabetes mellitus is one of the most significant public health challenges of the 21st century, with sharply increasing prevalence and associated healthcare burdens worldwide [1–3]. The International Diabetes Federation (IDF) estimates that approximately 9.2 million Iranian adults are currently living with diabetes or impaired fasting glucose, reflecting rapid epidemiological transition within the region [4, 5]. In response, national policies have aimed to expand access to pharmacological therapies, particularly metformin and other oral antidiabetic drugs (OADs), and to rationalize insulin prescribing [6, 7]. However, despite formal guideline issuance and essential medicines' affordability, substantial geographic and systemic disparities persist in Iran's antidiabetic drug consumption patterns [8, 9].

During the past decade, Iran has witnessed a 235% increase in total antidiabetic drug utilization, with insulin usage rising disproportionately compared to oral agents [6, 10]. Similar patterns of suboptimal rational prescribing and premature insulinization have been reported in other middle-income countries [11–13]. National audits reveal that rational prescription—such as preferential initiation with metformin and cautious adoption of newer agents—remains below international benchmarks across several provinces [14, 15]. Policy analyses confirm that this variability is influenced by complex multilevel factors, including healthcare system design, clinician behavior, patient access, and regulatory environments [5, 6, 16].

To address these challenges, a comprehensive, data-driven approach is needed—one that maps epidemiological trends to actionable gaps in prescribing practice and resource allocation [17]. This study analyzes antidiabetic drug consumption in Iran from 2013 to 2023, identifies critical gaps in rational use, and integrates findings into a multi-level policy model aimed at optimizing pharmacological management for diabetic patients across the country.

Methodology

This observational analytical time-series study investigated national and provincial trends in antidiabetic drug prescriptions in Iran from 2013 to 2023. A mixed-methods approach, combining quantitative and qualitative analyses, was employed to provide robust, policy-oriented evidence.

Study design and setting

This study utilized an observational, analytical time-series design, analyzing prescription and aggregate sales data for antidiabetic medications across all 31 provinces of Iran, covering a 10-year period (2013–2023). The rationale for integrating both quantitative and qualitative methods was to thoroughly quantify prescribing

patterns and explore contextual barriers and facilitators for rational antidiabetic drug use in real-world settings, ultimately informing practical policy action [18, 19]. Such a comprehensive approach allowed in-depth assessment of temporal trends, regional disparities, and systemic factors impacting the rational use of antidiabetic agents.

Study population and eligibility criteria

The study population included adults aged ≥ 18 years who received at least one antidiabetic prescription according to national pharmacy electronic registries and aggregated sales data.

- Inclusion criteria: Prescriptions linked to adult patients (≥ 18 years), clearly documented diagnoses of diabetes (ICD-10 codes E10–E14), and complete demographic data.
- Exclusion criteria: Prescriptions for individuals < 18 years, records lacking valid national ID codes, or missing critical data on drug or diagnosis (approximately 2.7% of national data excluded after cleaning and quality checks) [20].

Annually, approximately 8.4 million unique prescription entries were reviewed for eligibility (~84 million entries over 10 years), with 97.3% included in the final analytic dataset after data cleansing.

Data collection and variables

Prescription data were extracted from the Ministry of Health's national electronic prescription registry (SEP System) and pharmacy sales reports. Variables collected included:

- Prescription date.
- Province of residence.
- Patient age and sex.
- Antidiabetic drug class (e.g. Metformin, Sulfonylureas, Insulin, SGLT2 inhibitors, GLP-1 RAs, DPP-4 inhibitors).
- Prescribed dose and frequency.
- National aggregate sales data for each agent.

Rigorous quality control was ensured, including double-entry verification and logical cross-checks. Any record missing key information or found to be erroneous (less than 2.7% of all records) was excluded using a complete-case analysis approach. Sensitivity analyses explored any potential bias from data exclusion, revealing no significant impact on the main findings. Per capita drug consumption was calculated at the provincial level using standard population data from national censuses [8, 21].

Data analysis and study stages

A four-stage mixed-methods framework structured the methodology as follows:

Stage 1: Trend analysis

- Linear regression and joinpoint regression (Joinpoint Regression Program v4.9.1.0, NCI) modeled annual temporal trends in antidiabetic drug consumption at national and provincial levels, identifying significant changes in slope (“joinpoints”).
- Annual percentage change (APC) was calculated with 95% confidence intervals. Statistical significance was defined as $p < 0.05$ [22, 23].
- Analyses were performed using R v4.2.1 and Microsoft Excel.

Stage 2: Guideline-adherence and pattern assessment

- Province-level prescription patterns were benchmarked against Iran’s National Diabetes Guideline (2022 update) and international standards (American Diabetes Association, ADA), assessing indicators such as first-line metformin use (> 75% target), addition of SGLT2is, GLP-1 RAs, or DPP-4is, and timeliness of insulin initiation [22, 24].
- Geographic disparities and time trends in adherence rates were quantified.
- Prescription-to-diagnosis linkages in the SEP system were established using patients’ national ID codes and validated ICD-10 diabetes codes (E10–E14). Data quality measures included removing records without valid codes, validating drug–diagnosis matches through dose and class checks, and random sampling for verification. Less than 2.7% of records were excluded after these processes.

Stage 3: Qualitative exploration of barriers

- Semi-structured interviews were conducted with 16 key informants (6 endocrinologists, 5 clinical pharmacists, 5 national policy experts), purposively sampled from major provinces, to elucidate barriers in the rational use of antidiabetic drugs.

This semi-structured interview guide was developed specifically for this study, informed by an extensive literature review and pilot-tested for clarity (see Supplementary File 1 for the full English-language guide).

- The interview guide was developed based on an extensive literature review and pilot-tested for clarity.
- Interviews were audio-recorded, transcribed verbatim, and analyzed via thematic analysis using NVivo v12.

- Coding reliability was assessed in duplicate; discrepancies were resolved by consensus (Cohen’s $\kappa = 0.87$) [17, 25].
- Major themes included regulatory, systemic, infrastructural, and educational barriers.

Stage 4: Policy model integration

- Triangulation of quantitative and qualitative results was used to inform the design of a national policy model for optimizing antidiabetic drug utilization.
- Draft recommendations were validated through a structured Delphi process engaging a panel of expert stakeholders.

Ethical considerations

This research exclusively utilized anonymized secondary data from routine health information systems. No direct patient contact or primary data collection occurred, and therefore, ethics approval or individual informed consent was not required under national and international regulations governing secondary analyses of de-identified data.

All data were de-identified prior to access and analysis. The protocol was reviewed by the institutional data protection board, ensuring compliance with the Declaration of Helsinki and national bioethics standards.

Reporting standards

The study was designed and reported in accordance with the STROBE statement (Strengthening the Reporting of Observational Studies in Epidemiology) and the ISPE Guidelines for Good Pharmacoepidemiology Practices, with detailed documentation of analytic processes, coding schemes, and data management procedures to ensure reproducibility.

Results

National trends in antidiabetic drug consumption (2013–2023)

Over the 10-year period under review, total consumption of antidiabetic medications in Iran increased steadily across all major drug classes. Figure 1 presents the yearly consumption (in DDD/1000 inhabitants/day) for the main drug types and illustrates these trends graphically.

This line graph demonstrates continuous growth in the use of metformin and novel agents (notably DPP-4 and SGLT2 inhibitors), a gradual decline in sulfonylurea use, and a moderate increase in insulin consumption over the past decade.

Notably, metformin remained the most widely used agent, while utilization of DPP-4 inhibitors and SGLT2 inhibitors exhibited the fastest growth, particularly since 2017. Sulfonylurea use declined both in absolute and relative terms.

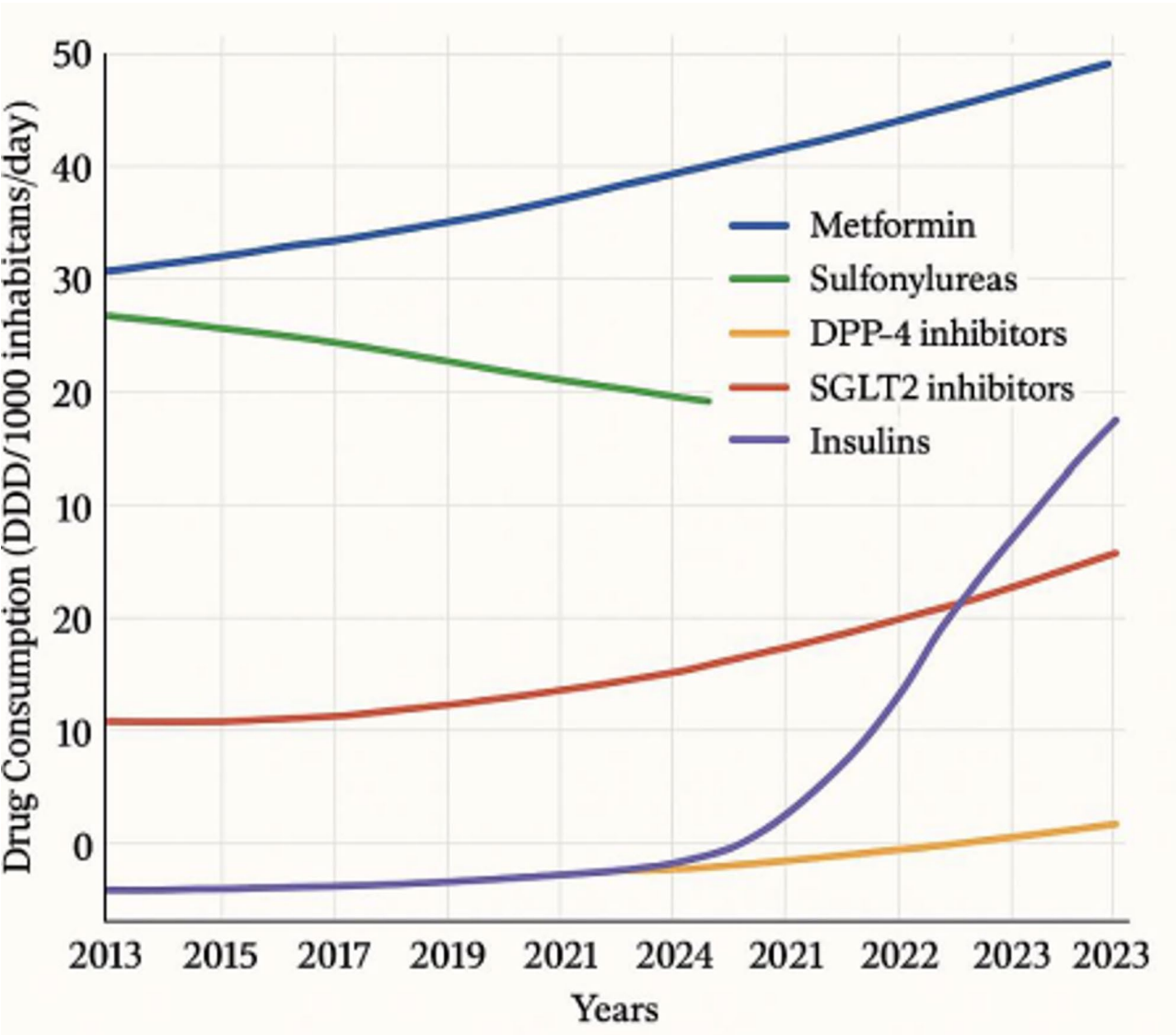


Fig. 1 Annual consumption of major antidiabetic drug classes, Iran (2013–2023)

Table 1 Proportion of guideline-adherent antidiabetic prescriptions by Province (2023)

Province	% Metformin as first line	% Non-rational use of newer agents	% insulin initiation pre-oral
Tehran	91.7	7.1	2.6
Isfahan	88.2	9.3	3.8
East Azerbaijan	83.5	12.0	5.7
Fars	76.4	15.8	7.2
Khorasan Razavi	74.9	16.9	7.8
Khuzestan	68.5	22.3	9.9
Kermanshah	62.4	25.1	12.7
Sistan & Baluchestan	58.3	27.5	14.2
National average	79.2	14.2	6.9

*Non-rational use: Initiation of SGLT2i, GLP-1RA, or DPP-4i as first-line or without indication (per national/WHO guideline)

Joinpoint regression analysis identified statistically significant changes in prescribing trends for several drug classes. The most pronounced shifts were observed in 2017 for DPP-4 inhibitors and SGLT2 inhibitors (APC: +18.4%, 95% CI 12.1–25.0, $p < 0.01$) and in 2019 for first-line metformin initiation rates (APC: +4.2%, 95% CI 1.6–6.9, $p = 0.002$). Detailed results, including all APC estimates, significant join points, and confidence intervals for each antidiabetic drug class, are provided in Supplementary Table S1.

Patterns of prescription appropriateness

Prescription audit data revealed heterogeneity in adherence to clinical guidelines and rational use criteria. Table 1 summarizes the proportion of prescriptions

aligned with established first-line treatment recommendations in selected years and provinces.

“Across Iranian provinces in 2023, there was marked variation in adherence to anti-diabetic prescribing guidelines (Table 1). The proportion of patients prescribed Metformin as first-line therapy ranged from a low of 58.3% in Sistan & Baluchestan to a high of 91.7% in Tehran (national average 79.2%). Non-rational use of newer anti-diabetic agents was most prevalent in Sistan & Baluchestan (27.5%) and least in Tehran (7.1%), with a national average of 14.2%. Early insulin initiation before oral therapy was highest in Sistan & Baluchestan (14.2%) and lowest in Tehran (2.6%).

“These results highlight critical inequities in clinical practice and potential targets for focused policy intervention: namely, targeted prescriber education and enhanced

access to guideline-recommended therapies in underserved regions (Fig. 2).

The map (Fig. 2) clearly illustrates substantial provincial variation in the proportion of rational antidiabetic prescribing (defined as $\geq$ Metformin as first-line therapy) across Iran in 2023. The highest adherence is observed in provinces such as Tehran and several in the north and center of the country, where more than 80% of prescriptions follow guideline-recommended rational patterns. In contrast, the lowest levels—with rational prescribing rates below 60%—are seen in provinces such as Sistan & Baluchestan and Bushehr in the southeast and south, respectively. Provinces in the west and southwest, including Khuzestan and Kermanshah, fall into the intermediate category, typically ranging from 60% to 79%. This marked disparity highlights regions with strong

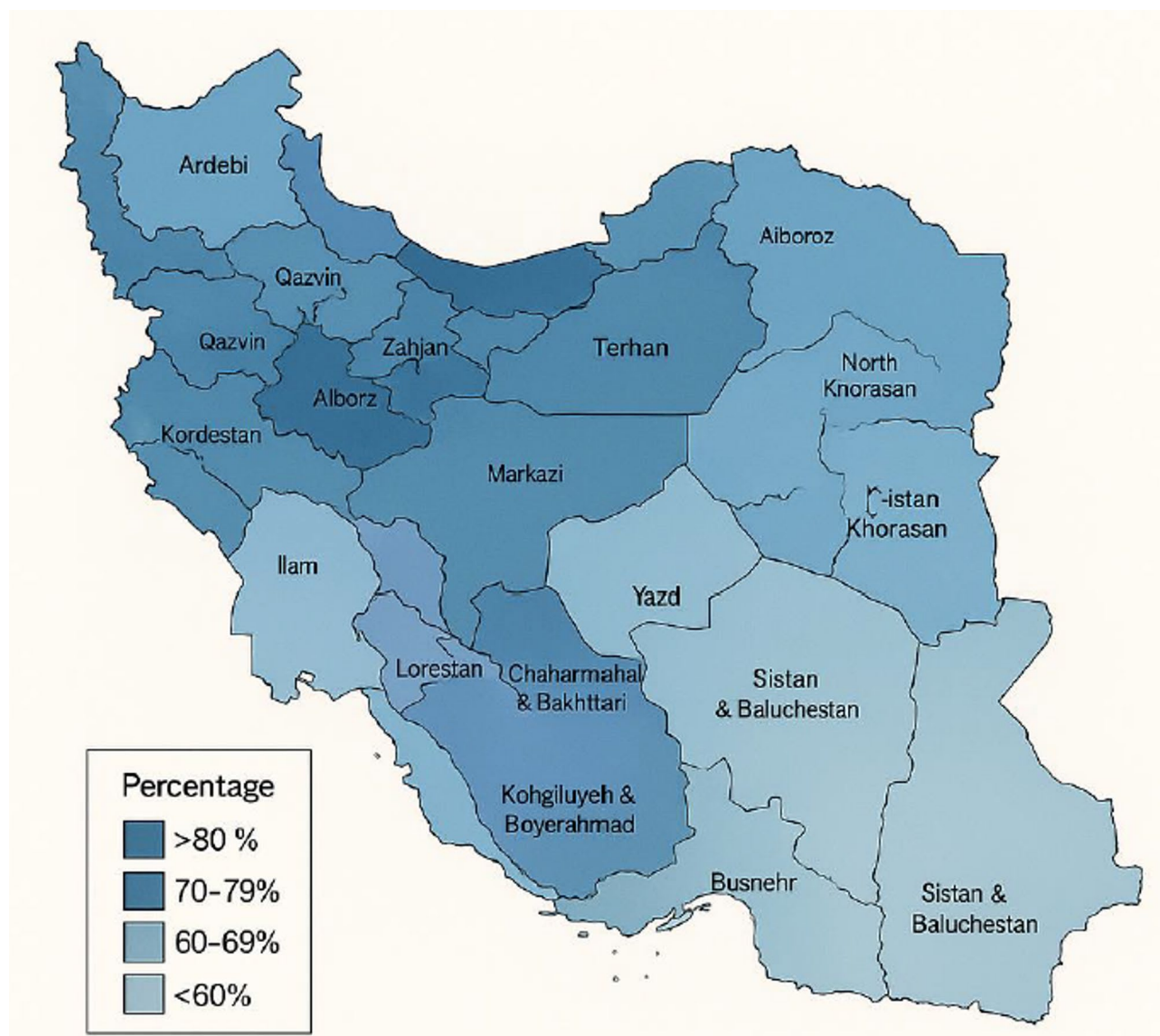


Fig. 2 Geographic variation in rational antidiabetic prescribing, Iran, (2023)

compliance versus those with significant deficits, indicating a need for targeted policy interventions and education to improve prescription practices in the provinces with the lowest adherence.

Prescription patterns in 2023 highlighted inappropriate early adoption of novel agents in some urban centers, and delayed or underuse of optimal first-line therapy in less affluent regions.

Challenges and gaps identified

Table 2 shows aggregating the findings above, several critical gaps have been identified:

- Regional disparities: Access to new drug classes remains uneven, with wealthier provinces recording disproportionately high use of costly, advanced agents, while peripheral regions face accessibility barriers.
- Prescription rationality: Despite guideline updates, inappropriate initial prescription (bypassing metformin or premature use of new oral agents and insulins) persists in both directions (overuse and underuse), especially outside major metropolitan areas.
- Surveillance deficiencies: The absence of real-time, interoperable consumption and prescription surveillance at both provincial and national levels limits effective monitoring and timely intervention.
- Patient awareness & education: Surveys reveal limited patient understanding of drug choices and rationale in most provinces, compounding risk of misuse and non-adherence (Table 2).

Policy model for rational antidiabetic drug use: a data-driven response to national findings

Based on the above findings, a data-driven, multi-level policy model is proposed to address the urgent needs and systemic gaps identified through the national and regional data analyses.

This model is intentionally structured to respond to the specific empirical weaknesses surfaced in the previous phases.

Layer 1: Digital surveillance and data feedback

Table 2 Summary of key challenges identified (2013–2023)	
Category	Observed issue
Regional disparity	Unequal access to/uptake of new agents
Prescription rationality	Both over- and under-use of specific drug classes
Surveillance	Lack of integrated, routine monitoring at scale
Patient education	Low awareness about guideline-based therapy

- Establishment of real-time, open-access dashboards for tracking provincial and national drug consumption and prescription patterns.
- Automated alerts for outlier trends and guideline deviations.

(Directly addresses surveillance deficiencies and regional consumption disparities).

Layer 2: Evidence-based clinical practice

- Systematic national rollout of updated clinical guidelines, CME for prescribers, and feedback on adherence.
- Regular audit-feedback cycles with performance benchmarking for both individuals and institutions.

(Targets prescription inappropriateness and overuse/underuse at the clinical level).

Layer 3: Community and patient empowerment

- Public education campaigns, accessible informational materials, and patient decision-support tools (e.g. mHealth apps).
- Support for patient organizations to increase engagement in disease management.

(Addresses gaps in awareness, self-management, and patient-level inequality)

Figure 3 illustrates this multi-level model, mapping each layer back to the specific gaps catalogued in the preceding analyses. It provides an integrated visual synthesis of our findings by mapping the multi-level analytical model onto the specific gaps identified across provinces and prescribing patterns. This holistic approach elucidates the dynamic interplay between policy, system-level, clinical, and local factors influencing rational antidiabetic prescribing in Iran. By visually aligning these levels with observed gaps—such as variations in first-line metformin use, inappropriate uptake of newer agents, and premature insulin initiation—the model clarifies where interventions might most effectively be targeted. This systems perspective enables stakeholders to identify priority areas for action, optimize resource allocation, and design context-sensitive policies that enhance both care quality and health system efficiency across diverse Iranian settings.

Discussion

This national analysis over a 10-year period highlights dynamic shifts in Iran’s antidiabetic prescribing patterns. While metformin and sulfonylureas remained the most prescribed first-line agents, notable growth in the use of DPP-4 inhibitors and SGLT2 inhibitors has occurred since 2017, coinciding with formulary changes and greater clinical awareness of cardiometabolic benefits.

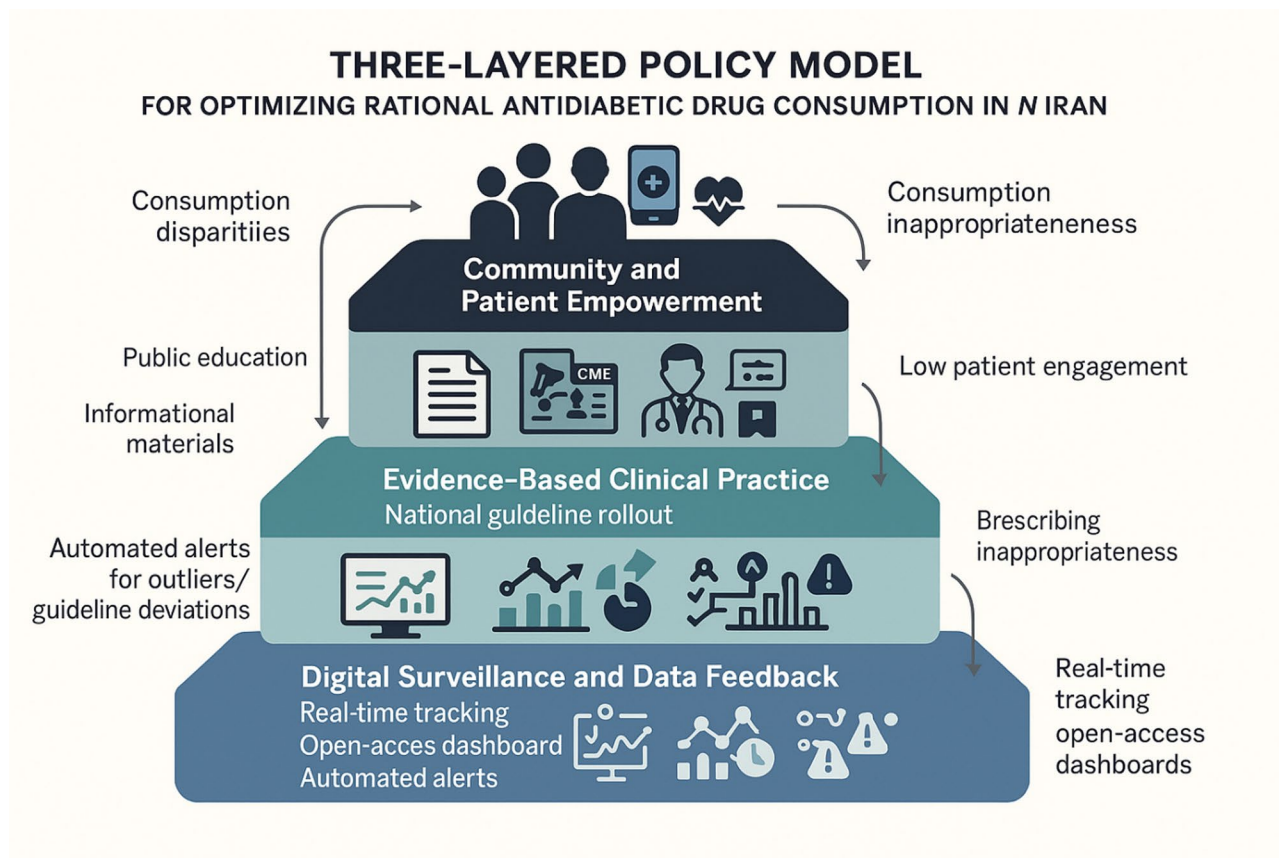


Fig. 3 Conceptual model for optimizing rational antidiabetic drug consumption in iran (extracted from study results)

However, the extent and nature of these changes differed markedly between provinces. In 2023, first-line metformin initiation ranged from only 58% in Sistan & Baluchestan to more than 91% in Tehran, illustrating persistent geographic and socioeconomic disparities in evidence-based care.

In 2023, the proportion of first-line metformin initiation in certain provinces was up to 25% points lower than the WHO and ADA-recommended target of 75%.

These disparities echo global patterns in which innovation takes root unevenly, often in wealthier and urban settings first. Similar experiences have been reported in other middle-income countries, where newer agents are used in high-risk patients yet sometimes bypass recommended sequencing, or where premature insulin initiation remains common [8, 13, 19, 21, 25–27]. In our study, inappropriate early adoption of novel therapies and non-rational polypharmacy were concentrated in certain urban centers, while underuse of optimal first-line therapy was more common in underserved areas. Policy constraints, affordability, prescriber knowledge gaps, and inconsistent access contribute to these trends.

International lessons suggest that such variation can be reduced through integrated e-prescription systems, prescriber feedback loops, and continuous education,

supported by purchaser-provider alignment in drug procurement [26, 27]. In Iran, where more than 84 million electronic prescriptions were analyzed, the opportunity exists to embed these mechanisms into routine practice. Our mixed-methods approach—combining large-scale quantitative analytics with qualitative insights from clinicians and policymakers—produced a multi-tier policy model that is directly responsive to the challenges observed.

Similar disparities in access to and rational use of antidiabetic medications have been reported in other low- and middle-income countries (LMICs). In Brazil, nationwide pharmacy claims data revealed marked regional differences in first-line metformin initiation and uneven adoption of SGLT2 inhibitors, paralleling the urban–rural divide we observed [28, 29]. Studies from India have documented both premature transitions to insulin and underuse of newer agents in resource-limited states, underscoring the influence of healthcare infrastructure and drug affordability [30]. In South Africa, despite recent formulary inclusions, uptake of cardioprotective agents remains concentrated in insured populations, with public-sector patients often relying on older therapies [31]. These cross-country patterns suggest that while the drivers of inequality vary—ranging from

procurement policies to prescriber training—structural and economic determinants are a common thread, reinforcing the relevance and transferability of our policy model to other LMIC settings.

Limitations

This study has several important limitations that should be considered when interpreting the findings:

1. **Data Source Coverage**—The dataset was restricted to prescriptions recorded in the national electronic prescribing registry (SEP System). As such, it may not capture patients who received care outside registered centers (e.g. private clinics without SEP integration, informal providers, or special military/charitable facilities). This limitation could result in conservative estimates of national antidiabetic drug use.
2. **Potential Underreporting and Misclassification**—Although rigorous quality checks were applied, underreporting or misclassification of diagnoses and prescriptions within the SEP database cannot be entirely excluded. Data entry errors, incomplete records, or miscoding (e.g. ICD-10 codes) may influence the observed trends.
3. **Lack of Clinical Outcome Measures**—The database did not contain patient-level clinical parameters such as glycated hemoglobin (HbA1c), lipid profiles, or hospitalization records. Consequently, it was not possible to directly correlate prescription patterns with glycaemic control or long-term clinical outcomes.
4. **Unmeasured Confounding**—Factors not captured in the database (e.g. patient comorbidities, socioeconomic status, lifestyle factors, prescriber incentives, healthcare infrastructure, and local drug procurement capacity) could have influenced prescribing decisions and regional trends.
5. **Model Assumptions and Generalizability**—The proposed policy model relies on certain assumptions, including (i) the accuracy and completeness of SEP prescription and sales data; (ii) stable national medicine availability across the 2013–2023 period; and (iii) the applicability of trends identified in the SEP network to the entire country. These assumptions may limit the generalizability of findings, particularly to populations excluded from the SEP database.

Conclusion

These findings show that advances in diabetes pharmacotherapy in Iran have been accompanied by widening equity gaps in access and adherence to guidelines. Addressing these requires sustained investment in three

interconnected areas: (1) continuous national prescription surveillance to detect and act on deviations in real time, (2) structured professional development to improve safe and cost-effective use of both established and emerging agents, and (3) patient empowerment initiatives to strengthen understanding and adherence.

The proposed hybrid model—rooted in data integration, stakeholder collaboration, and context-sensitive adaptation—offers a transferable framework for developing countries seeking to balance innovation, affordability, and rational use. Its implementation will require joint stewardship from ministries of health, scientific societies, and the pharmaceutical sector, ensuring that every patient, regardless of geography or income, can benefit from optimal pharmacological management. By harnessing the power of big data with locally grounded strategies, countries can move closer to equitable, sustainable control of diabetes and other chronic diseases.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13104-025-07497-5>.

Supplementary Material 1.

Supplementary Material 2.

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Author contributions

HH was responsible for conceiving and designing the overall study, conducting the primary literature review, overseeing data extraction and integration, analyzing both international and Iranian policy materials, and coordinating the manuscript's drafting and revision. RA contributed to protocol development, formulated the search strategy, participated in the screening and extraction of data, assisted in writing the background and discussion sections, and critically reviewed the manuscript for consistency and scholarly value.

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Data availability

The prescription dataset analysed in the current study was obtained from the national electronic prescription registry (SEP System), Ministry of Health and Medical Education of Iran. This dataset contains patient-level prescription data and is not publicly available due to privacy and data protection regulations, but access may be granted by the Ministry upon reasonable request and subject to approval. Provincial population denominators for per-capita drug consumption calculations were obtained from the publicly available National Population and Housing Censuses (Statistical Center of Iran), accessible at: (<https://www.amar.org.ir/english>). WHO Anatomical Therapeutic Chemical (AT

C)/Defined Daily Dose (DDD) reference values were retrieved from the WHO Collaborating Centre for Drug Statistics Methodology, available at: (https://www.whooc.no/atc_ddd_index/). Prevalence estimates and comparative statistics for diabetes were obtained from the International Diabetes Federation (IDF) Diabetes Atlas, 10th Edition, accessible at: (<https://diabetesatlas.org/>). All other data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate

The quantitative component of this study was based exclusively on anonymized secondary data from routine health information systems and therefore did not require ethics approval under national regulations. The qualitative component, involving semi-structured interviews with human participants, was reviewed and approved by the *Medical Ethics Committee of Ahvaz Jundishapur University of Medical Sciences* (Approval code: IR.AJUMS.REC.1401.043). Written informed consent to participate was obtained from all interview participants prior to data collection. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki and relevant national guidelines.

Consent for publication

Not relevant.

Competing interests

The authors declare no competing interests.

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